

Interoperability of Information Systems in the Ecuadorian Public Sector

Josue Merino Calderon¹, Sebastián Arias², Nahir Carrera³, and Nicolás Suquillo⁴

Department of Computer Science, Universidad de las Fuerzas Armadas ESPE,
Sangolquí, Ecuador.

{ejmerino, xsarias, ndcarrera2, fnsuquillo}@espe.edu.ec

Abstract. Public administration in Ecuador is undergoing a process of change, moving away from isolated systems toward an articulated digital environment. This descriptive and exploratory research addresses the structure of the Governmental Interoperability Platform (PIG), emphasizing its technological core: the Governmental Service Bus (BSG). It analyzes how the interconnection of databases from key institutions, such as the Civil Registry and the Internal Revenue Service (SRI), has contributed to a 77% reduction in the need for in-person procedures at the national level. Finally, it is determined that, despite having a solid technological base, semantic interoperability and the adequate protection of personal data remain the main challenges for consolidating the digital transformation of the State.

Keywords: Interoperability · Governmental Service Bus · Digital Government · Civil Registry · SRI · Administrative Simplification.

1 Introduction

For decades, red tape in Ecuador has constituted a significant barrier to the efficiency of the state apparatus and the country’s competitiveness. Until a few years ago, the task of articulating public information fell on the citizen, who had to move physical documentation between institutions operating in isolation. This situation generated not only a considerable economic impact —quantified in millions of dollars annually— but also a progressive deterioration of trust in public entities [1].

The enactment of the *Organic Law for the Optimization and Efficiency of Administrative Procedures* in 2018 marked a turning point, turning interoperability from a technical best practice into a legal obligation. Specifically, Art. 9 of this regulation establishes the prohibition of requesting information that already exists in state records, forcing public institutions to integrate into a centralized data exchange node. In this context, this research analyzes the structure of said node and evaluates its impact on the reduction of in-person attendance in administrative procedures.

2 Methodology

This research employs a qualitative approach with a descriptive and exploratory scope. A systematic documentary review of the current legal regulations in Ecuador and the technical reports issued by the Ministry of Telecommunications (MINTEL) and the National Directorate of Public Registries (DINARP) between 2018 and 2025 was carried out.

To evaluate the status of the services, a comparative analysis was applied based on the digital maturity levels proposed by international organizations, categorizing data provider institutions according to their response capacity and integration with the Governmental Service Bus (BSG).

3 State of the Art and Legal Framework

Interoperability in the public sector has ceased to be a technical peripheral and has become the core of New Public Management (NPM). At a global level, the undisputed benchmark is Estonia's *X-Road* model, a distributed data exchange architecture that guarantees data sovereignty and security through end-to-end encryption [9]. In the Ibero-American context, this trend has materialized in the *Ibero-American Charter of Electronic Government*, promoted by CLAD, which advocates for the citizen's right not to provide documents already held by the public administration [10].

Ecuador, aligned with these regional trends, has moved from a model of information islands, where institutions like the Internal Revenue Service (SRI) and the Civil Registry operated in technological silos, toward an integrated ecosystem. This evolution is based on four regulatory and doctrinal milestones that define the current state of the art in the country:

1. **Ministerial Agreement 005-2018 (Technical Interoperability Standard):** This document is the technical pillar that establishes the four levels of interoperability adapted from the European Interoperability Framework (EIF). It highlights semantic interoperability, which requires institutions to use common data dictionaries so that the concept of address or family size is identical for the Ministry of Economic and Social Inclusion (MIES) as it is for the SRI [3].
2. **Organic Law for the Optimization and Efficiency of Administrative Procedures (2018):** Represents a legal paradigm shift by introducing the *Principle of Expediency* and the *Presumption of Truthfulness*. Its greatest impact is Art. 9, which prohibits the demand for physical copies of IDs and voting certificates, forcing direct consultation with the Governmental Service Bus (BSG). This legal milestone reduced the bureaucratic burden by eliminating the citizen as the State's messenger [2].
3. **Organic Law on Personal Data Protection (2021):** Although not often mentioned as an interoperability milestone, this law is fundamental to the current state of the art. It establishes that the exchange of information

between, for example, the Civil Registry and a public banking entity must have a legal basis or explicit consent, introducing the concept of "secure and ethical interoperability" [11].

4. **Organic Law on Digital and Audiovisual Transformation (2023):**

This is the most recent milestone that grants full legal validity to the digital file and the electronic signature. This law allows notary and registry processes, which historically required in-person attendance due to the "solemnity of paper," to be integrated into the interoperability flow, enabling the reduction of procedures in sectors previously considered impassable.

The state of the art reveals that the current trend is shifting toward *proactive interoperability*, where the State does not wait for the citizen to start a procedure but, through real-time data cross-referencing (governmental Big Data), identifies needs and offers services automatically—a level of maturity that the SRI has already begun to implement in suggested tax returns [12].

4 Technical Architecture: The Service Bus (BSG)

The Governmental Service Bus (BSG) is an Enterprise Service Bus (ESB) type solution that facilitates information exchange under a Service Oriented Architecture (SOA). This *middleware* managed by MINTEL allows institutions to operate as data Providers or Consumers.

4.1 Protocols and Standards

To ensure security and integrity, the BSG uses standardized protocols:

- **SOAP/WSDL:** For legacy web services that require high security and transactionality.
- **REST/JSON:** For modern and agile integrations that facilitate connection with mobile applications and the *gob.ec* portal.
- **Authentication:** Use of tokens and digital certificates to ensure that only authorized entities access sensitive information.

5 Case Studies: Pillars of Interoperability

The effectiveness of the platform depends on the quality of the data from its main providers.

5.1 General Directorate of Civil Registry (DIGERCIC)

The Civil Registry provides the "Identity Data Validation Service." Before interoperability, a citizen had to present a physical copy of their ID. Today, institutions like the Ministry of Health or public universities consume a web service that returns the photograph, marital status, and birth data in real time, reducing fraud by identity theft and eliminating paper.

5.2 Internal Revenue Service (SRI)

The SRI is the largest provider of financial and tax data. Its interoperability allows for:

- **RUC Validation:** Instant verification of the active status of taxpayers.
- **Tax Clearance Certificates:** Elimination of the need to visit agencies to obtain a physical stamp; it is now validated through data cross-referencing in public procurement processes [5].

6 Evaluation Criteria

To determine the impact of interoperability on public management, this research adopts the Digital Government Maturity Model proposed by regional organizations such as ECLAC [6]. This framework allows institutions to be categorized not only by their network presence but by their data exchange capacity. The maturity levels defined for this study are:

1. **Informative:** The entity offers static information and procedure requirements but does not allow online management.
2. **Interactive:** Allows the downloading of forms and basic communication, although the citizen is still responsible for moving the documentation.
3. **Transactional:** The procedure is completed on digital portals (such as *gob.ec*), but the manual uploading of digitized documents from other entities persists.
4. **Transformative (Interoperable):** Represents the maximum stage of maturity where, through the Governmental Service Bus (BSG), institutions exchange data in real time, eliminating the need for the administered party to present physical requirements.

7 Analysis of Results and Impact

The application of the maturity criteria reveals an asymmetric evolution in the Ecuadorian public sector. The results indicate that interoperability has had a resounding success in entities with highly standardized databases, while sectors with decentralized processes show a significant lag.

Table 1 presents a qualitative synthesis of the analyzed sectors, evaluating their level of integration with the BSG and the consequent reduction in the use of physical media.

As observed, the Internal Revenue Service (SRI) and the General Directorate of Civil Registry have reached a transformative state, where the exchange of information is "transparent" to the user. This has allowed critical processes, such as identity validation or tax solvency, to operate under the "once-only" principle, where the State does not request data it already possesses. Conversely, in the municipal sphere, the lack of a unified Service Oriented Architecture (SOA) limits the impact of interoperability, maintaining dependence on in-person validation in the territory.

Table 1. Maturity levels and impact on the reduction of in-person attendance.

Sector	Maturity Level	Paper Reduction	Lead Entity
Fiscal / Tax	Transformative	Very High	SRI
Identity and Civil	Transformative	High	Civil Registry
Higher Education	Transactional	High	SENESCYT
Public Health	Transactional	Medium	MSP / IESS
Municipal Management	Interactive / Initial	Limited	Municipal GADs

8 Discussion and Challenges

Despite the fact that the technical deployment of the Governmental Service Bus (BSG) has proven to be a robust and scalable infrastructure, the research reveals that the true friction in the Ecuadorian digital ecosystem does not lie in the communication protocols, but in the so-called organizational interoperability. This phenomenon manifests in a cultural resistance rooted in what various authors call the "bureaucracy of the stamp" [14].

8.1 Cultural Resistance and Fear of Audits

Paradoxically, the existence of proactive regulations such as the Law for the Optimization of Procedures clashes directly with internal control processes. Many public servants, conditioned by the rigor of the State Comptroller General's Office, maintain a systemic distrust toward digital records. There is a perception that data obtained via interoperability lacks the necessary "solemnity" to support an administrative process, which leads to the redundant demand for "certified" or "notarized" physical documents. This dichotomy generates an inefficient hybrid system where the web service is consumed, but the paper is archived, doubling the operational load instead of reducing it.

8.2 The LOPDP Bottleneck

A critical finding of this study is the collateral impact of the *Organic Law on Personal Data Protection (2021)*. While the regulation is a necessary advance for the citizen's digital sovereignty, its implementation has generated a "paralysis by analysis" in institutional legal counsel.

Currently, opening a new web service in the BSG does not depend on technical feasibility, but on an exhaustive legal justification that is often interpreted restrictively. The fear of sanctions by the Data Protection Superintendency has led provider institutions, such as the Civil Registry or the SRI, to impose access barriers under the argument of "data minimization," slowing down the integration of cross-cutting procedures that could be resolved in seconds.

8.3 Technological Asymmetry in Local Governments

While the Executive Branch shows interoperability rates above 70%, the Decentralized Autonomous Governments (GAD) operate in a fragmented reality. The

lack of minimum standards in the development of their municipal *Core Systems* prevents many of them from acting as consumer nodes of the BSG. This technical asymmetry creates an "Ecuador at two speeds": a citizen can file taxes with the SRI in minutes from their mobile, but must go in person to a small municipality to deliver a physical copy of that same tax payment, breaking the digital value chain.

8.4 The Challenge of Semantic Interoperability

Finally, the challenge of data semantics arises. It is not enough for systems to connect; it is imperative that they understand the same thing. During the investigation, inconsistencies were detected in how different entities define critical variables such as "address," "dependents," or "disability." Without a unified national data ontology, technical interoperability runs the risk of propagating systemic errors, where erroneous data at the primary source is automatically replicated throughout the entire state apparatus without clear citizen rectification mechanisms.

9 Conclusions

The analysis developed allows for the conclusion that the interoperability ecosystem in Ecuador has moved beyond the technical experimentation stage to consolidate itself as the cross-cutting axis of modern public management. The implementation of the Governmental Service Bus (BSG) has transformed the state architecture, allowing neurological institutions such as the Internal Revenue Service (SRI) and the General Directorate of Civil Registry to operate as high-availability provider nodes. This progress has shifted the burden of data integration from the citizen to the technological infrastructure, drastically reducing the need for in-person interactions.

However, the transition to a fully digital administration model under the premise of "Zero Paper" does not depend solely on the availability of *Web Services*, but on the resolution of structural gaps. In this sense, the following priority lines of action are established:

1. **Reduction of institutional technological asymmetry:** it is imperative to standardize the technical capabilities of smaller-scale Decentralized Autonomous Governments (GAD). National interoperability will only be effective when municipal systems can consume and provide data at the same pace as the Executive Branch, preventing citizens from encountering islands of physical bureaucracy at the local level.
2. **Evolution toward digital trust and cultural change:** the end of the "bureaucracy of the stamp" requires a pedagogical effort oriented toward public servants and control bodies. It is necessary to decouple the legal validity of an administrative act from its physical support, strengthening the use of BSG digital traceability as sufficient evidence for audits and special examinations by the Comptroller's Office.

3. **Implementation of the "Data Sovereignty" model:** the future of the platform must converge toward a citizen-centered approach, where the citizen ceases to be a passive subject and assumes control over their information. This implies the development of interfaces where the administered party can authorize, in real time, the exchange of their sensitive data between entities, guaranteeing strict compliance with the Organic Law on Personal Data Protection without sacrificing procedural agility.

Ultimately, the success of interoperability in Ecuador will depend on the State's ability to harmonize technological efficiency with legal security, ensuring that the simplification of procedures is a guaranteed citizen right and not just an operational optimization goal.

References

1. Barragán Martínez, X. O.: Factores de incidencia del gobierno electrónico en la gestión pública del Ecuador. Tesis Doctoral, UASB, Quito (2019).
2. Asamblea Nacional del Ecuador: Ley Orgánica para la Optimización y Eficiencia de Trámites Administrativos. Registro Oficial 353 (2018).
3. Ministerio de Telecomunicaciones: Norma Técnica de Interoperabilidad Gubernamental. Acuerdo Ministerial 005-2018, Quito (2018).
4. MINTEL: El Gobierno Nacional logró digitalizar el 77,48% de trámites. Boletín Oficial 1849, Guayaquil (2021).
5. Servicio de Rentas Internas: Catálogo de Servicios Web de Interoperabilidad para el Sector Público. SRI, Quito (2023).
6. Naser, A.: Gobernanza digital e interoperabilidad gubernamental: una guía para su implementación. CEPAL, Santiago de Chile (2021).
7. DINARP: Informe de interoperabilidad y registros públicos del Ecuador. Dirección Nacional de Registros Públicos (2024).
8. Portal gob.ec: Estadística de Trámites en Línea. <https://www.gob.ec/estadisticas>, acceso febrero 2026.
9. Vassilakopoulou, P., Marmier, A.: The Estonian X-Road: A Case of Public Sector Interoperability. In: Digital Government: Concepts, Methodologies, Tools, and Applications, pp. 450–468. IGI Global (2022).
10. CLAD: Carta Iberoamericana de Gobierno Electrónico. Centro Latinoamericano de Administración para el Desarrollo, Caracas (2021).
11. Asamblea Nacional del Ecuador: Ley Orgánica de Protección de Datos Personales. Registro Oficial Suplemento 459, 26 de mayo de 2021.
12. Servicio de Rentas Internas: Estrategia de Transformación Digital y Facilitación Tributaria 2024-2026. SRI, Quito (2024).
13. United Nations: E-Government Survey 2024: Accelerating Digital Transformation for Sustainable Development. United Nations Department of Economic and Social Affairs (2024).
14. Barragán, X.: Factores de incidencia del gobierno electrónico en la gestión pública: El caso de Ecuador. Revista de Gestión Pública, Vol. VIII, No. 1, pp. 35–62 (2019).
15. Contraloría General del Estado: Manual de Auditoría de Tecnologías de la Información en el Sector Público. Dirección de Capacitación, Quito (2023).
16. MINTEL: Informe de Brecha Tecnológica y Madurez Digital en Gobiernos Autónomos Descentralizados. Ministerio de Telecomunicaciones (2025).

17. Zuleta, M.: La protección de datos como barrera u oportunidad en la interoperabilidad gubernamental. En: *Journal of Iberoamerican Digital Law*, vol. 12, pp. 88–104 (2024).